

Computed Properties

Beyond Read-Only

What are Computed Properties?

- Derived values that automatically update when dependencies change
- Cached – only re-computed when dependencies change
- Usually read-only... but they don't have to be!
- Can define both a getter AND a setter

Basic Computed (Read-Only)

```
<script setup lang="ts">
import { ref, computed } from 'vue'

const firstName = ref<string>('John')
const lastName = ref<string>('Doe')

const fullName = computed<string>(
  () => `${firstName.value} ${lastName.value}`
)
</script>

<template>
  <p>Full name: {{ fullName }}</p>
</template>
```

Writable Computed

```
<script setup lang="ts">
import { ref, computed } from 'vue'

const firstName = ref<string>('John')
const lastName = ref<string>('Doe')

const fullName = computed<string>({
  get: () => `${firstName.value} ${lastName.value}`,
  set: (newValue: string) => {
    const [first, ...rest] = newValue.split(' ')
    firstName.value = first
    lastName.value = rest.join(' ')
  }
})
</script>

<template>
  First: <input v-model="firstName" /><br/>
  Last: <input v-model="lastName" /><br />
  Full: <input v-model="fullName" />
</template>
```

Example

- Form field with formatted display
- Currency input that stores cents
- User edits formatted value
- Setter parses and stores raw value

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  get: () => `${firstName.value} ${lastName.value}`,
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    firstName.value = first
    lastName.value = rest.join(' ')
  }
})
</script>

<template>
  First: <input v-model="firstName" /><br/>
  Last: <input v-model="lastName" /><br />
  Full: <input v-model="fullName" />
</template>
```

First: John

Last: Doe

Full: John Doe

Questions?

When have you needed writable computed properties?

Thank You!

Remember: Computed properties can be more than just getters!